A Programmable-Oscillator-Based Frequency-Shift Keying Radio Frequency Transmitter for Wildlife Tracking

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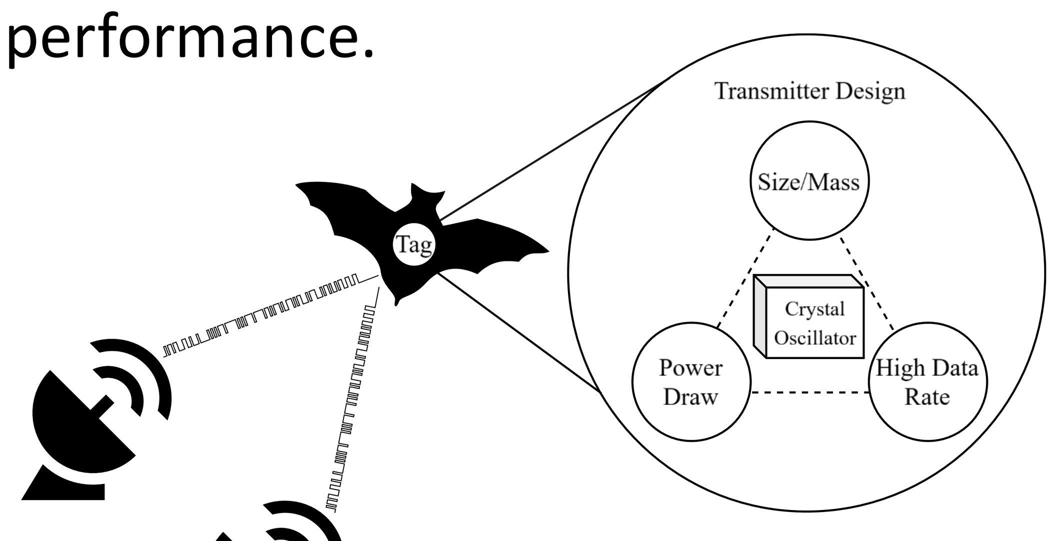
Introduction

Wildlife tracking helps study ecology, movement, and interactions with energy infrastructure. Time difference of arrival (TDOA) radio frequency (RF) telemetry tracking systems require modulation of a high chip-rate pseudorandom code for optimal performance.



Visualization of TDOA bat tag's baseband Gold code transmission and design constraints.

Objective

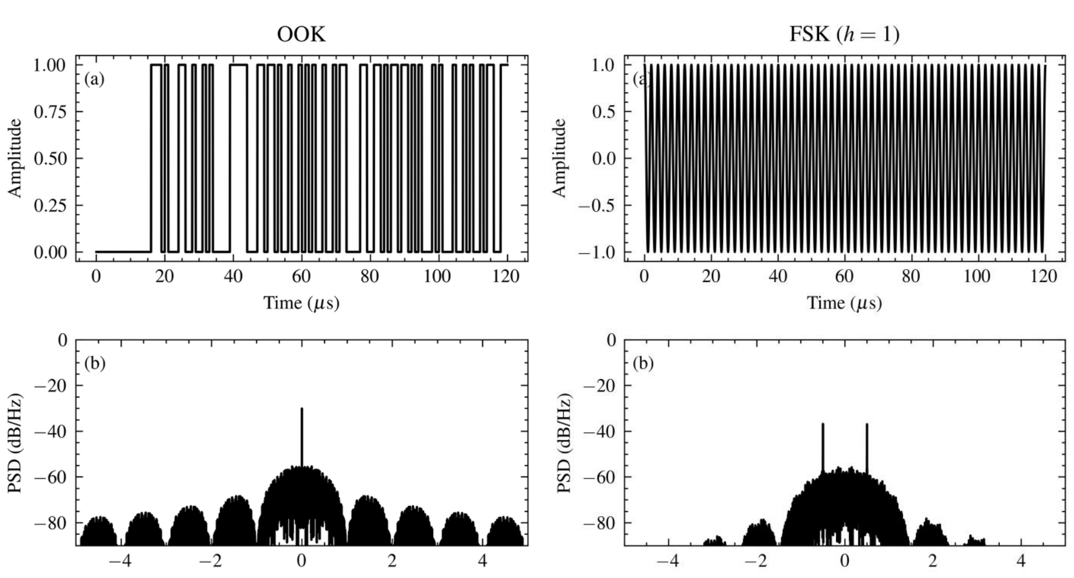
- Investigate the feasibility of using a programmable crystal oscillator architecture to generate frequency-shift keying (FSK) modulated, high chip-rate Gold Code.
- Characterize and compare performance and tradeoffs between different implementations.

Theory

Selected oscillators programmed at frequency ω generate a square wave oscillating between amplitudes A and *0*.

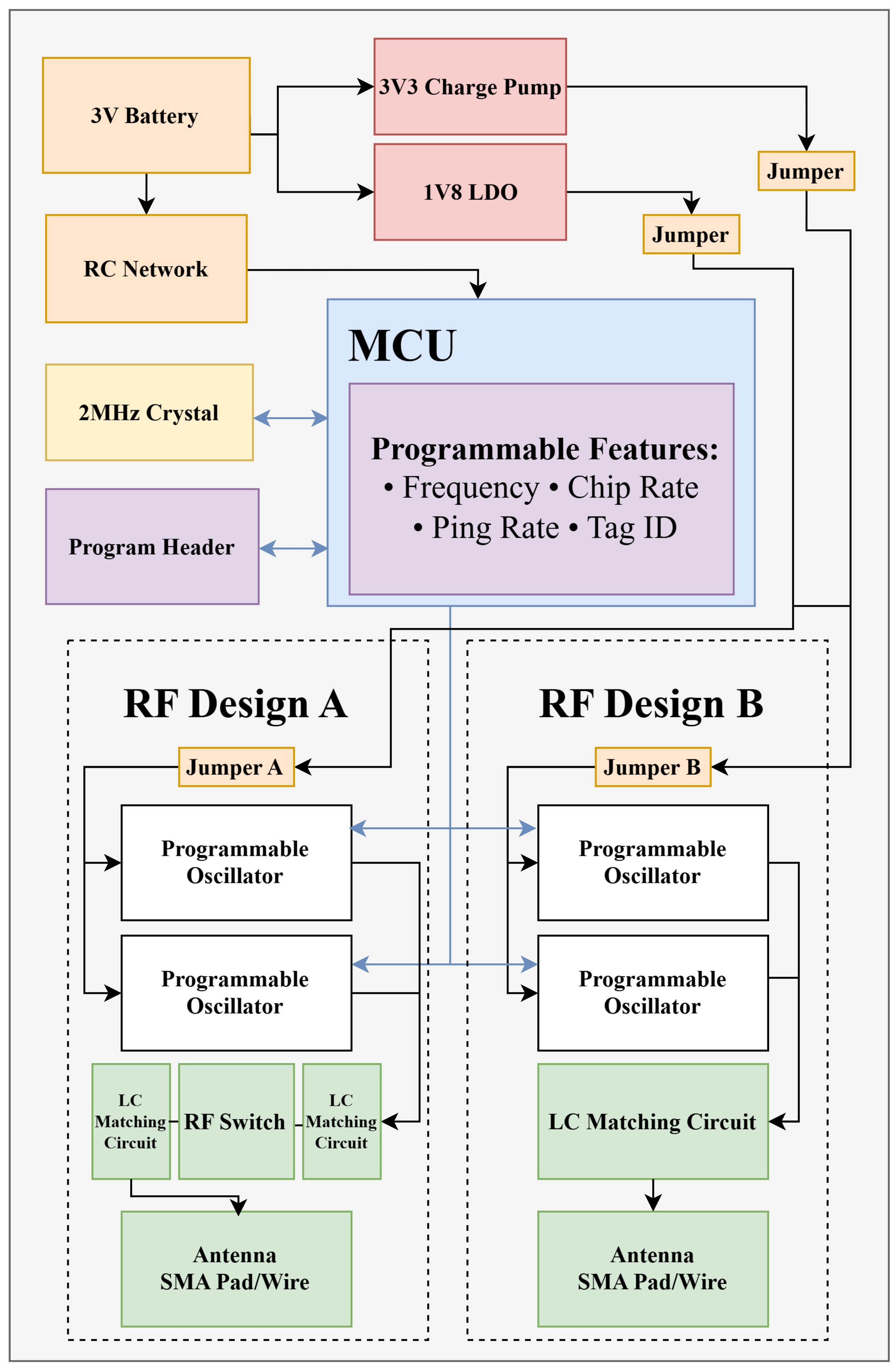
$$f(t) = \frac{A}{2} + \frac{2A}{\pi}\left(\sin(\omega t) + \frac{1}{3}\sin(3\omega t) + \frac{1}{5}\sin(5\omega t) + \cdots\right)$$

To modulate data onto these harmonics, the oscillators are toggled on/off at precise intervals.



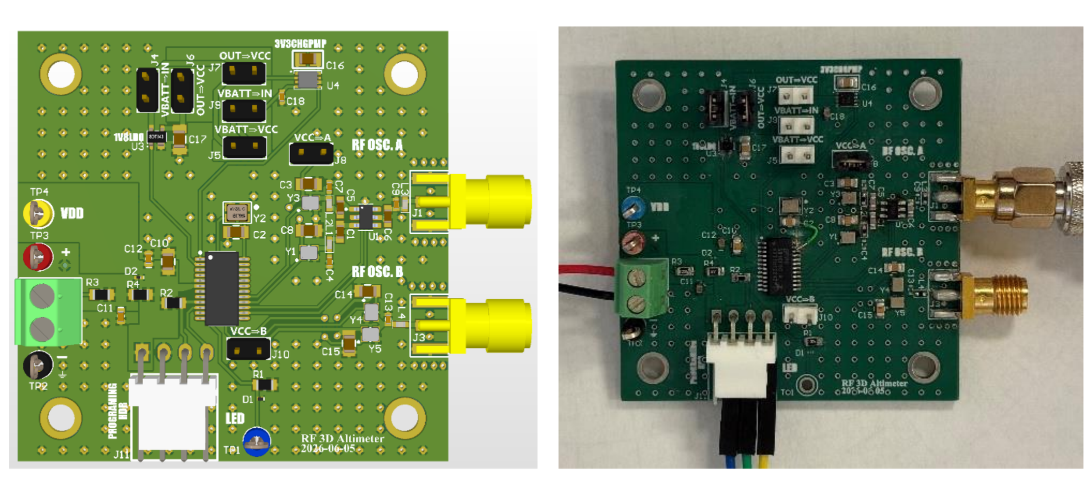
Baseband visualizations in time and frequency domain of Gold Code modulated using OOK or FSK.

Design



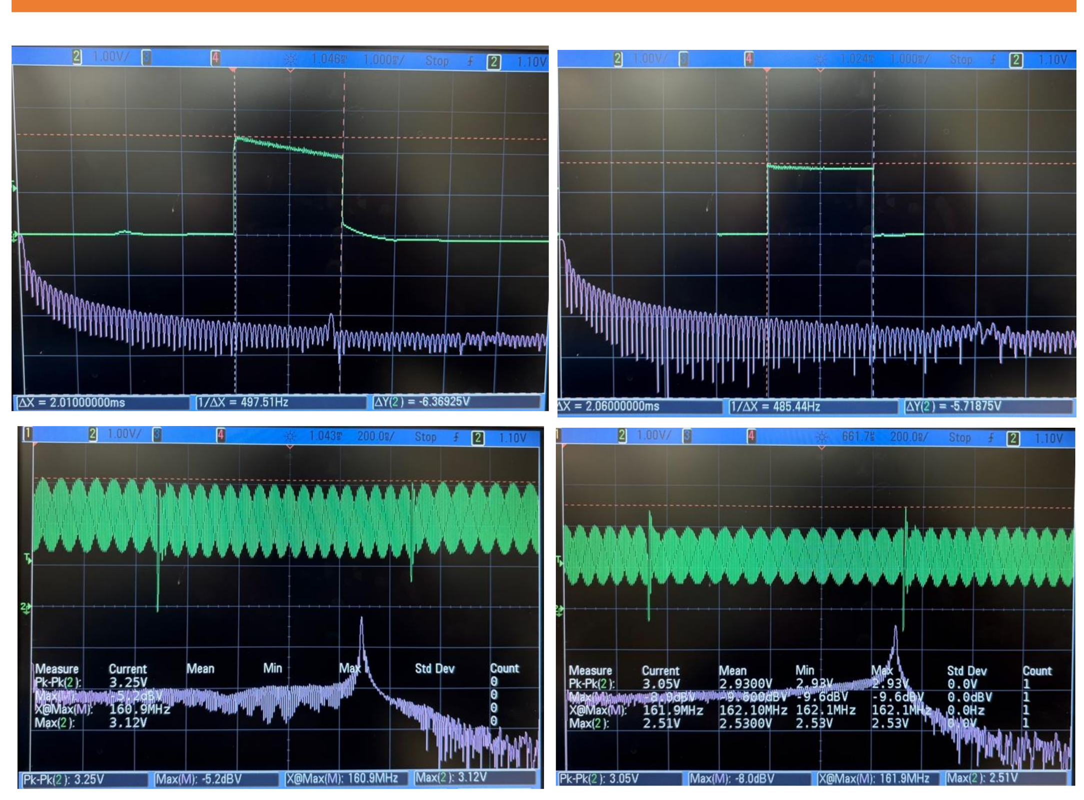
Circuit diagram of printed circuit board (PCB) with multiple test designs to evaluate the effects of RF switch on signal characteristics.

Implementation



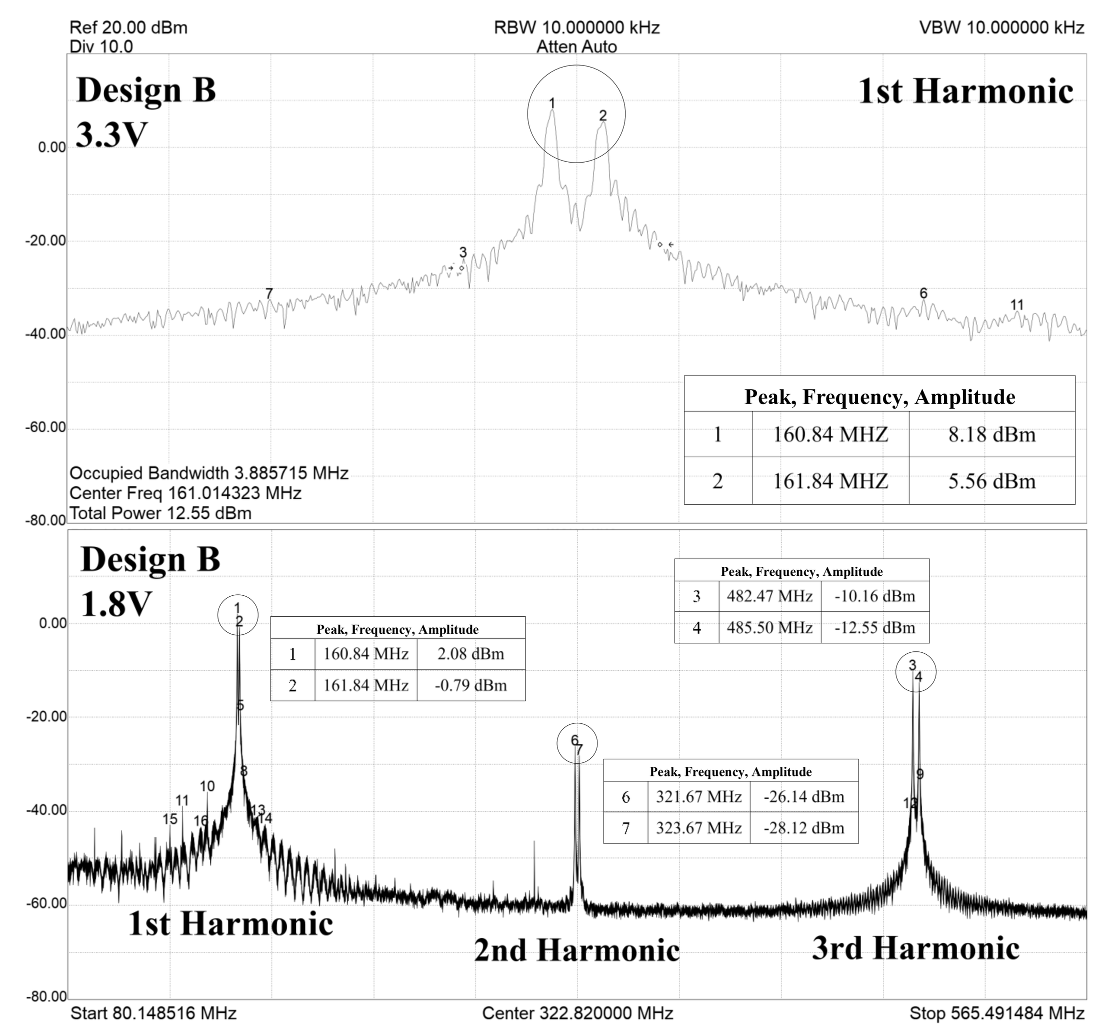
PCB in Altium versus PCB after assembly.

Results: Time Domain



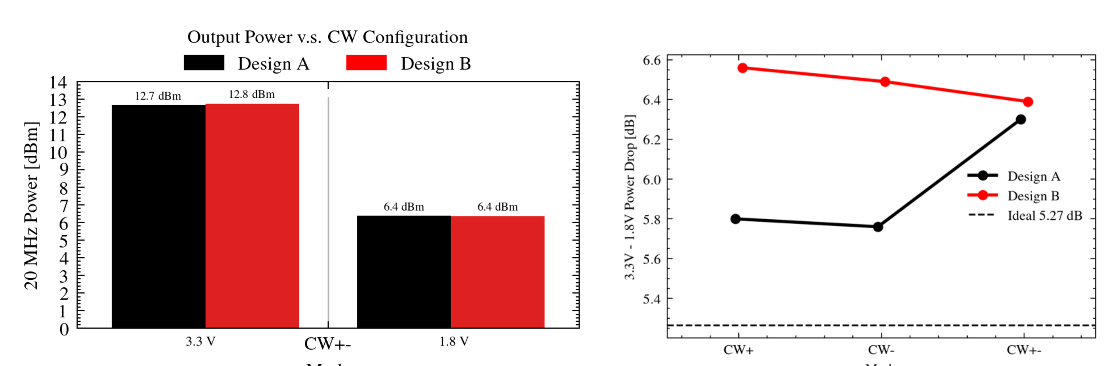
Top: Gold code burst at 1 ms/div. Bottom: Zoomed-in segment with frequency toggles and phase discontinuities at 200 ns/div. Left: Design A. Right: Design B.

Results: Freq. Domain



Outputted square wave harmonics from continuous-wave (CW) frequency toggling transmissions show two clear tones.

Results: Output Power



Left: Measured output power across 50 Ω load. (12.7±0.15) dBm at VCC = 3.3 V. (6.34±0.21) dBm at VCC = 1.8 V. Right: On average, there is a 6.21 dB power difference due to supply voltage drop.

Results: Life Span

System and VCC	Energy per transmission	Service life for 2 s ping rate
OOK 1.8 V 5th Harmonic[a]	116 μJ	15 days
OOK 1.8 V 3rd Harmonic[a]	149 μJ	12 days
FSK 3.3 V 1st Harmonic	251 μJ	6 days
ATLAS FSK 3.3 V[b]	737 μJ	2 days

Life spans for different supply voltages and operating harmonics.

Results: Field Tests

Preliminary tests @ Research Support Warehouse and Horn Rapids Park.

Conclusion

An FSK-enabled tag prototype using programmable oscillators for TDOA-based wildlife tracking was developed. The prototype supports transmitting at a 1 MHz chip rate and can be successfully detected at 342 m.

References

[a] J. Lu, H. Li, C. Tian, M.J. Myjak, J. Xiao, B.J. Bellgraph, S.S. Cartmell, and Z.D. Deng, "A Frequency-Programmable Miniaturized Radio Frequency Transmitter for Animal Tracking," Sensors 21(19), 6683 (2021).
[b] A.W. Weiser, Y. Orchan, R. Nathan, M. Charter, A.J. Weiss, and S. Toledo, "Characterizing the Accuracy of a Self-Synchronized Reverse-GPS Wildlife Localization System," in 2016 15th ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN), (IEEE, Vienna, Austria, 2016), pp. 1–12.

7/22/2026 | PNNL-SA-225246

Acknowledgements

This work was supported in part by the U.S. Department of Energy, Office of Science, Office of Workforce Development for Teachers and Scientists (WDTS) under the Science Undergraduate Laboratory Internships (SULI) Program.

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